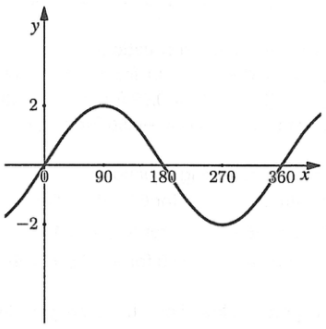
Graph Sketching: sin(x), cos(x), tan(x) in Degrees

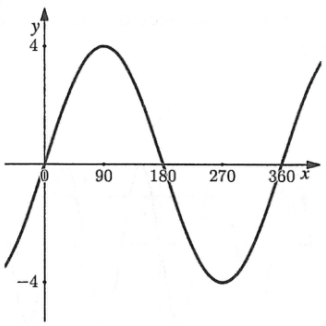
Questions 1–25

Question 1. Find the equation of each of the following graphs in the form $y = a\sin x$.

a)

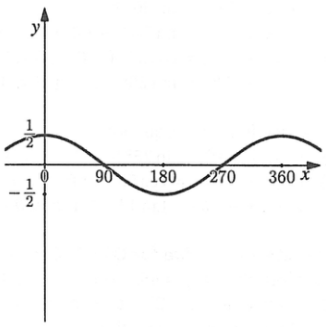


b)

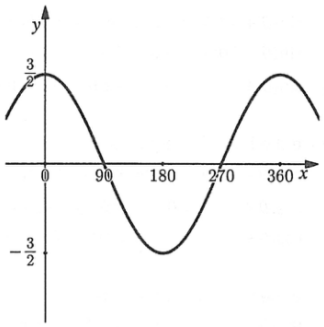


Question 2. Find the equation of each of the following graphs in the form $y = a\cos x$.

a)



b)



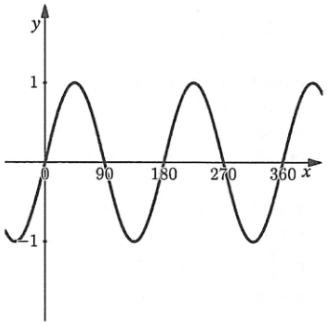
Question 3. Sketch the following graphs for $0° \leq x \leq 360°$:

a) $y = \frac{6}{5} \sin x$

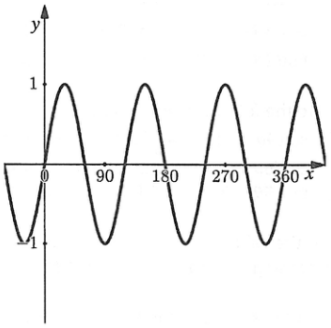
b) $y = \frac{2}{3} \cos x$

Question 4. Find the equation of each of the following graphs in the form $y = \sin ax$.

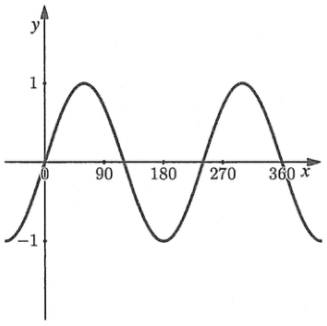
a)



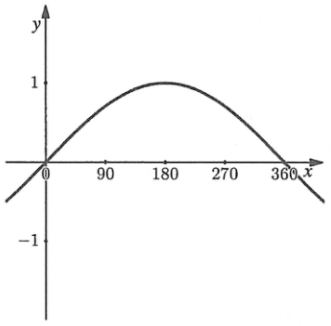
b)



c)



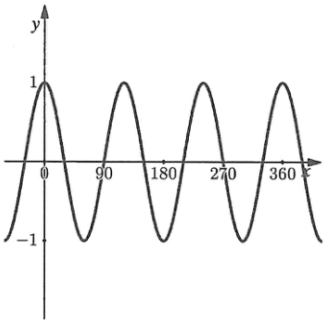
d)



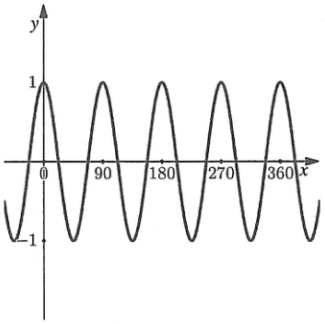
Question 5. Sketch the graph of $y = -2\sin x$ for $0° \leq x \leq 360°$.

Question 6. Find the equation of each of the following graphs in the form $y = \cos ax$.

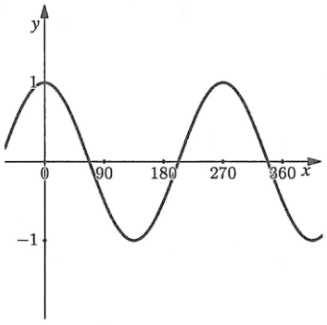
a)



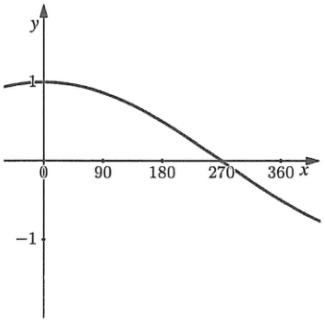
b)



c)



d)

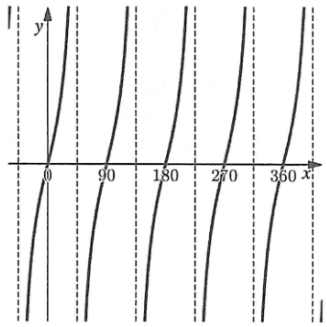


Question 7. Sketch the graph of $y = -3\cos x$ for $0° \leq x \leq 360°$.

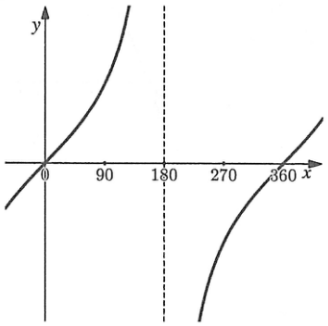
Question 8. Sketch the graph of $y = \sin(-2x)$ for $0° \leq x \leq 360°$.

Question 9. Find the equation of each of the following graphs in the form $y = \tan ax$.

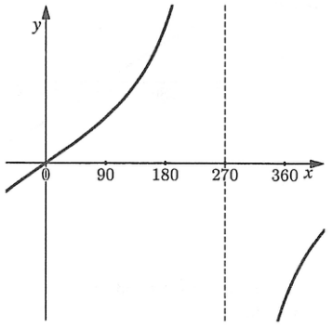
a)



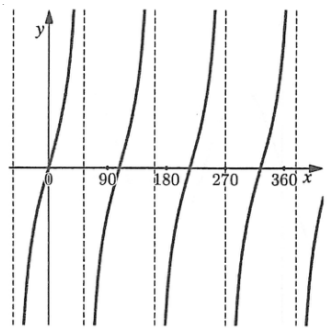
b)



c)



d)



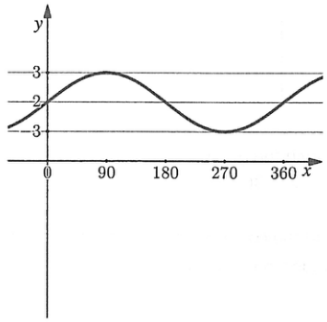
Question 10. Sketch the following graphs for $0° \leq x \leq 360°$:

a) $y = \tan \frac{3}{2}x$

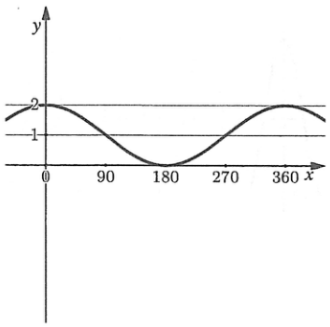
b) $y = \tan \frac{2}{3}x$

Question 11. Find the equation of each of the following graphs in either the form $y = a + \sin x$ or the form $y = a + \cos x$.

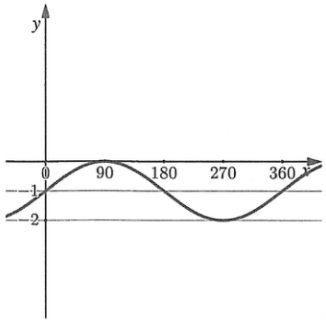
a)



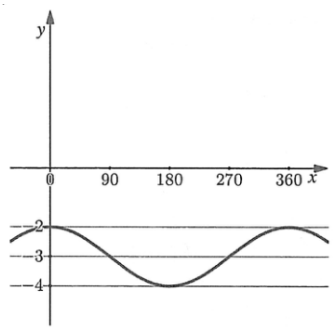
b)



c)

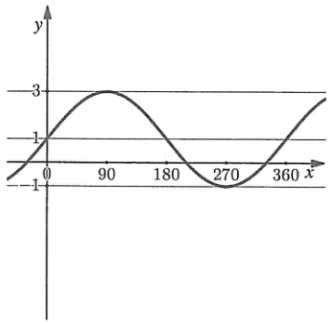


d)

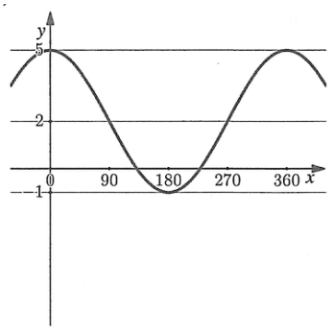


Question 12. Prerequisite knowledge for Q12–Q13: Combinations of Graph Transformations
Find the equation of each of the following graphs in the form $y = a + b\sin x$ or the form $y = a + b\cos x$.

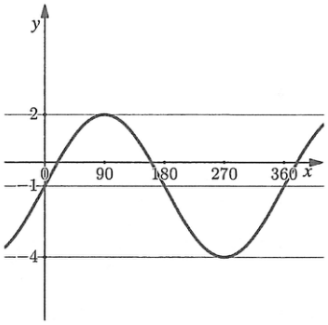
a)



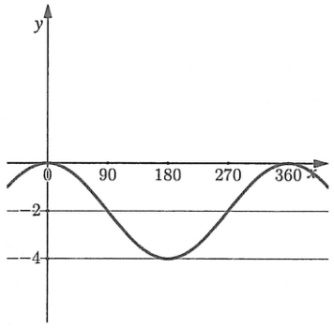
b)



c)



d)



Question 13. Sketch the following graphs for $0° \leq x \leq 360°$:

a) $y = 1 + 3\sin x$

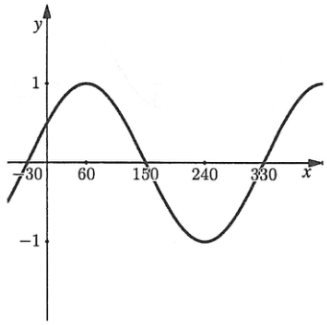
b) $y = -1 + 4\cos x$

c) $y = 2 - 2\sin x$

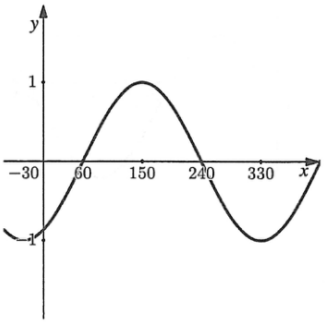
d) $y = -3 - 2\cos x$

Question 14. Find the equation of each of the following graphs in the form $y = \sin(x + a)$.

a)

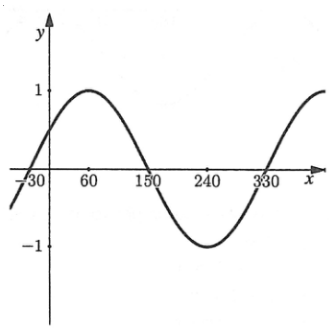


b)

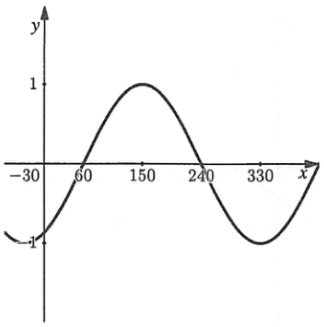


Question 15. Find the equation of each of the following graphs in the form $y = \cos(x + a)$.

a)

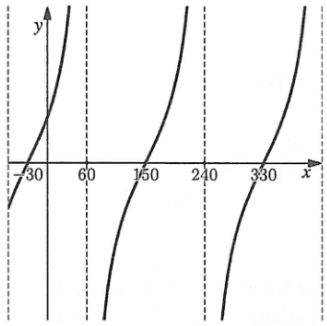


b)

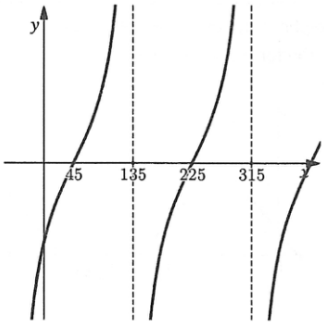


Question 16. Find the equation of each of the following graphs in the form $y = \tan(x + a)$.

a)



b)



Question 17. It is given that k is a constant where $k > 1$. Sketch the following graphs for $0° \leq x \leq 360°$:

a) $y = k + \sin x$

b) $y = -k + \cos x$

Question 18. Sketch the following graphs for $0° \leq x \leq 360°$:

a) $y = \sin(x - 90°)$

b) $y = \cos(x + 60°)$

c) $y = \tan(x + 90°)$

d) $y = \sin(x + 120°)$

e) $y = \cos(x - 40°)$

f) $y = \tan(x - 50°)$

Question 19. It is given that k is a constant where $0° < k < 90°$. Sketch the following graphs for $0° \leq x \leq 360°$:

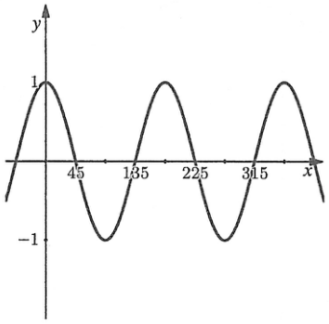
a) $y = \sin(x + k)$

b) $y = \cos(x - k)$

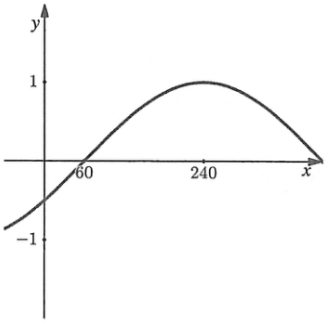
Question 20. Prerequisite knowledge for Q20–Q25: Combinations of Graph Transformations

Find the equation of each of the following graphs in the form $y = \sin(ax + b)$.

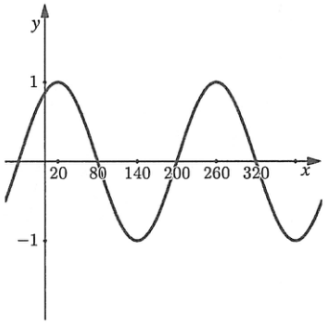
a)



b)

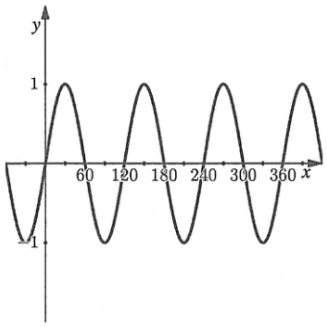


c)

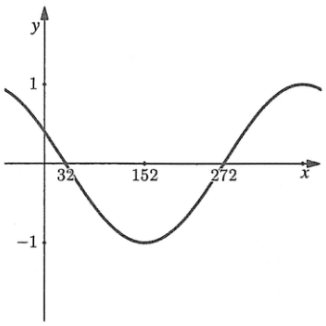


Question 21. Find the equation of each of the following graphs in the form $y = \cos(ax + b)$.

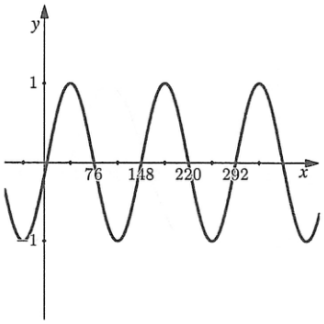
a)



b)

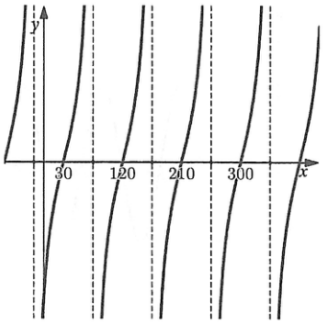


c)

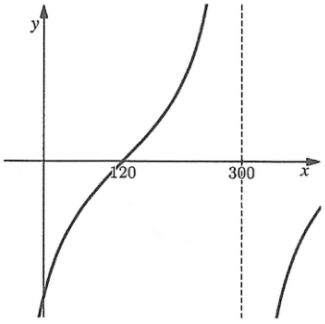


Question 22. Find the equation of each of the following graphs in the form $y = \tan(ax + b)$.

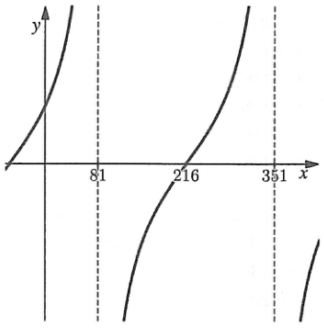
a)



b)

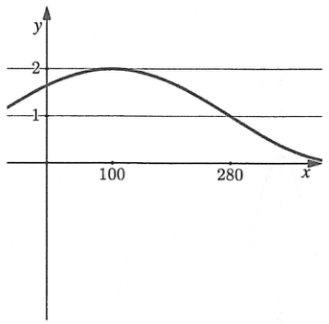


c)

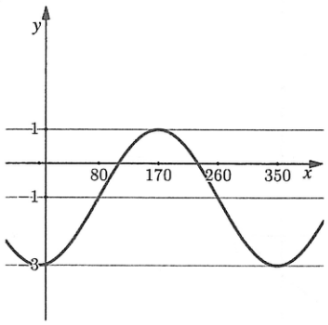


Question 23. Find the equation of each of the following graphs in the form $y = a + b\sin(cx + d)$ where $-180° < d < 180°$.

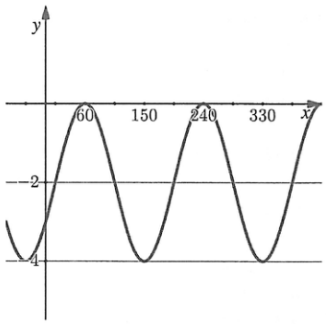
a)



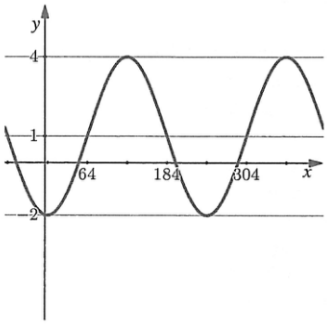
b)



c)

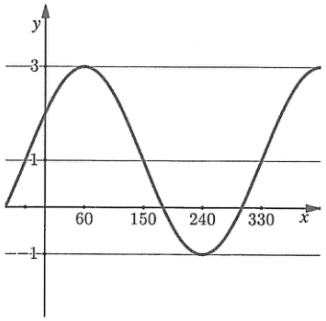


d)

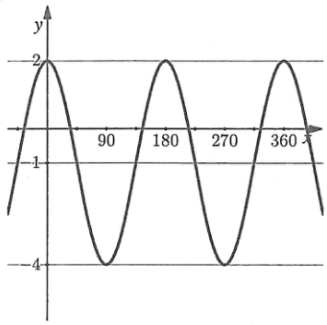


Question 24. Find the equation of each of the following graphs in the form $y = a + b\cos(cx + d)$ where $-180° < d < 180°$.

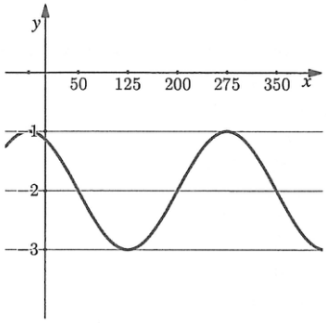
a)



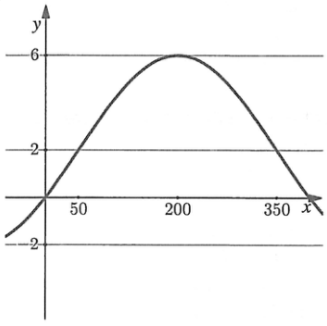
b)



c)



d)



Question 25. It is given that a and b are positive constants. Sketch the following graphs for $0° \leq x \leq 360°$:

a) $y = a + b\sin x$ where $a > b$

b) $y = a + b\sin x$ where $a = b$

c) $y = a + b\sin x$ where $a < b$